Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

AI PRINCIPLES AND PRACTICE · WEEK 8 OF 12

The Mathematics That Matters

Three simple ideas behind every model you have built

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Today, in one sentence

Three simple questions explain why the methods you have used actually work.

SHAPE	SURPRISE	SPACE
which way does the data stretch?	*how surprised is the model?*	*where does real data live?*
eigenvectors · SVD · low rank	**entropy · cross-entropy · KL**	**high dimensions · manifolds**

The maths was already there

Week 5 · PCA

It found the way your data stretches most.

Week 7 · the loss

It measured how surprised the model was by the right answer.

Week 9 · LoRA

It will change a huge model using two thin strips of numbers.

Today you see what is underneath each one.

PART 1

Shape

Which way does the data stretch?

YOU ARE HERE

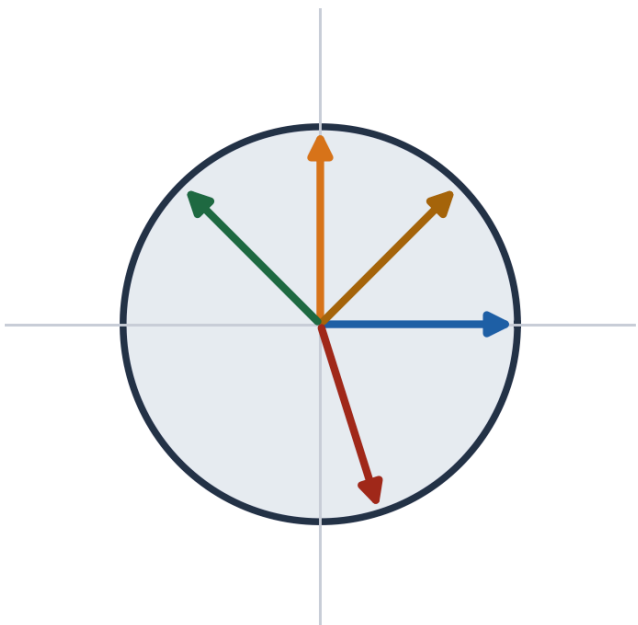
SHAPE
which way does the data stretch?

SURPRISE
how surprised is the model?

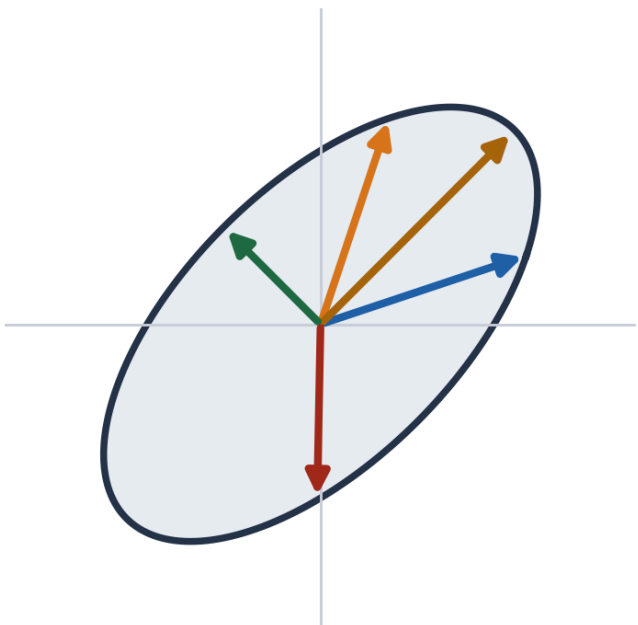
SPACE
where does real data live?

A table of numbers can move points

before: arrows on a circle



after multiplying by the matrix



Multiplying by a matrix stretches and turns every arrow.

Some directions only get longer



Like stretching a rubber sheet: arrows along the stretch keep their direction.

Eigenvectors, as a formula

THE FORMULA

$$Av = \lambda v$$

IN WORDS

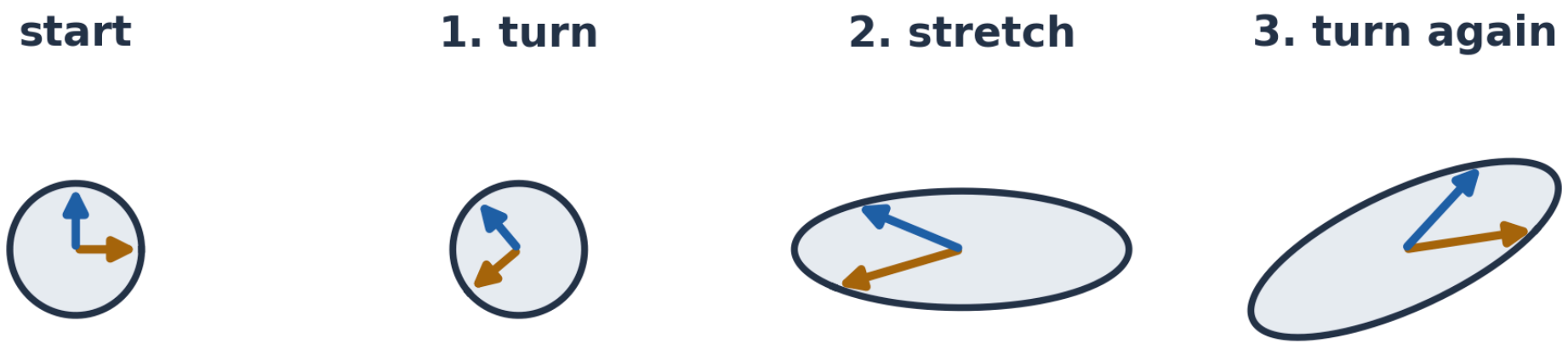
matrix × special direction = stretch factor × the same direction

WITH NUMBERS

[[3, 1], [1, 3]] · (1, 1) = (4, 4) = 4 · (1, 1)

The special direction is an **eigenvector**; its stretch factor λ (“lambda”) is an **eigenvalue**.

Every matrix does three simple things



Turn, stretch, turn. That is the **SVD** (singular value decomposition).

The SVD, as a formula

THE FORMULA

$$A = U\Sigma V^T$$

IN WORDS

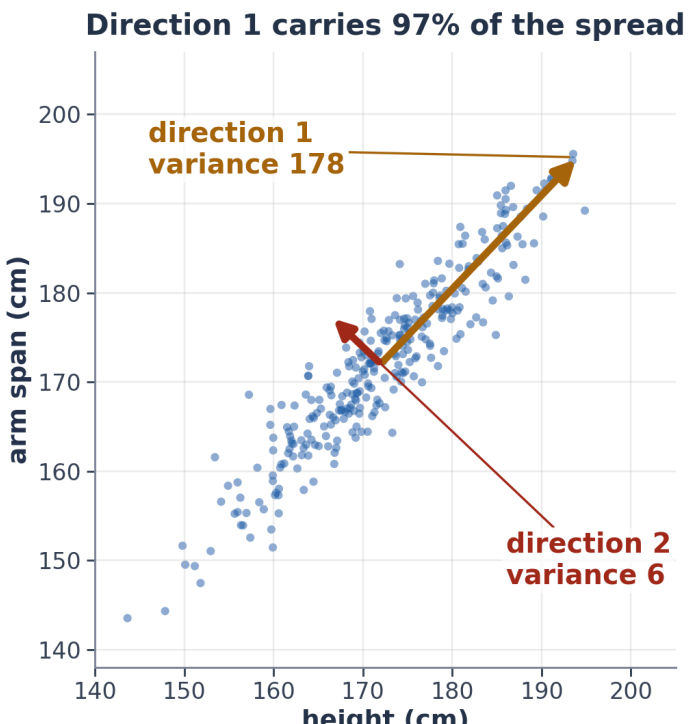
any matrix = turn × stretch × turn

WITH NUMBERS

the matrix above stretches by 2.54 one way and 0.88 the other

The stretches, $\sigma_1, \sigma_2, \ldots$ (“sigma”), are listed biggest first.

Which way does your data stretch most?



That longest direction is exactly what PCA finds.

PCA via SVD, as a formula

THE FORMULA

$$X_c = U\Sigma V^T$$

IN WORDS

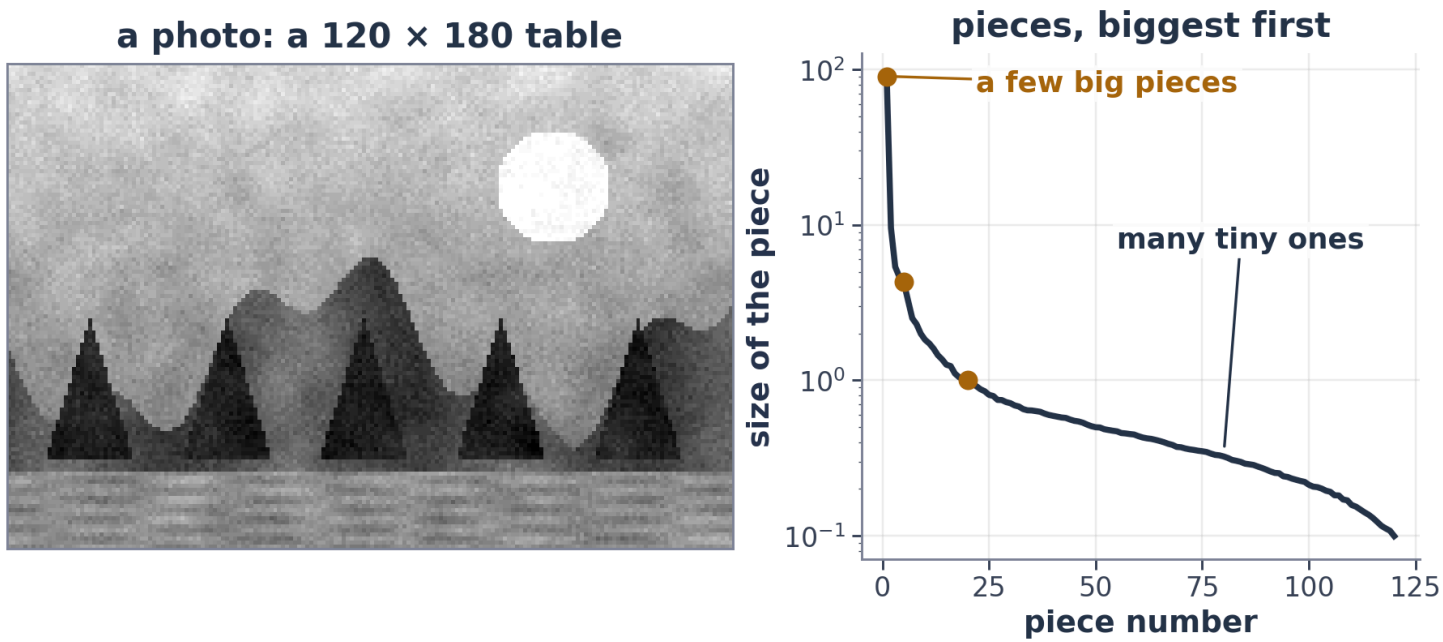
centred data = turn × stretch × turn; the biggest stretch is the main direction

WITH NUMBERS

height and arm span: 97% of the spread lies along one direction

Centred means you subtract each column’s average first. You write this yourself in today’s lab.

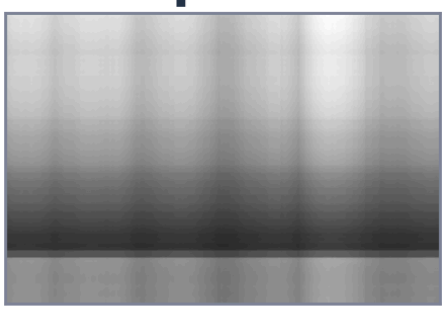
A photo is just a table of numbers



Its SVD splits it into pieces: a few big ones, many tiny ones.

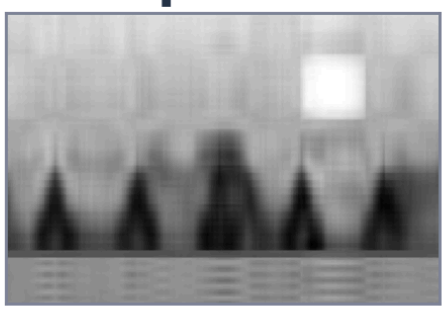
Keep the big pieces, drop the tiny ones

1 piece



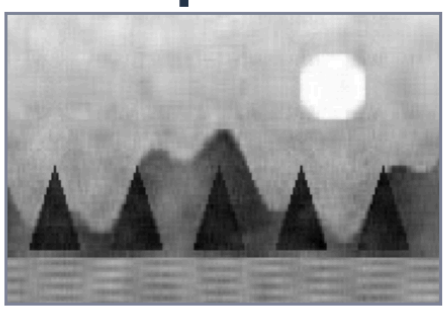
1% of the numbers
16.4% off

5 pieces



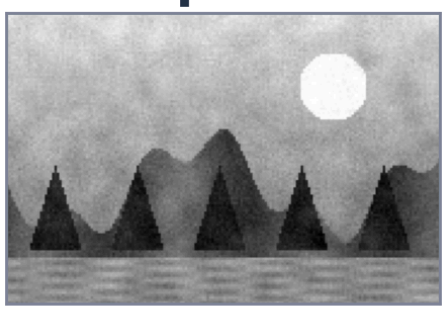
7% of the numbers
8.9% off

20 pieces



28% of the numbers
4.9% off

60 pieces



84% of the numbers
2.4% off

20 pieces look almost perfect, and use about a quarter of the numbers.

Low rank, as a formula

THE FORMULA

$$A \approx \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^T + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^T + \ldots + \sigma_k \boldsymbol{u}_k \boldsymbol{v}_k^T$$

IN WORDS

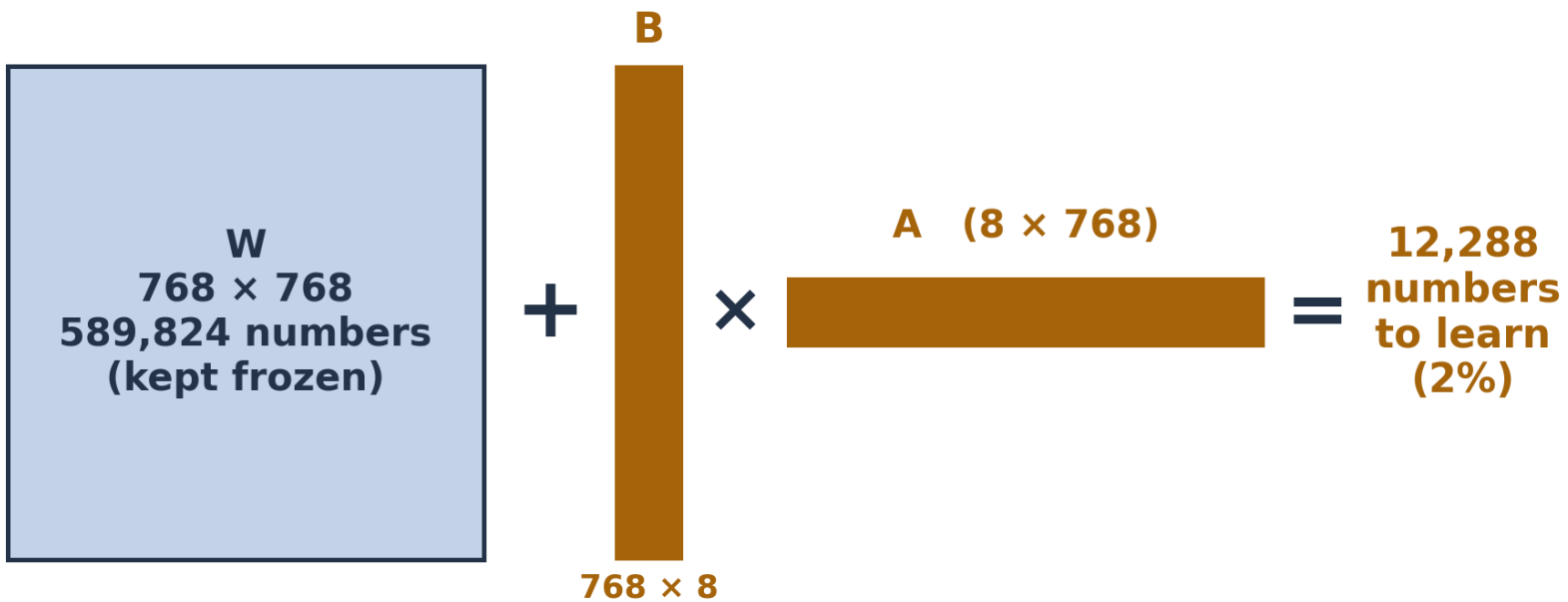
picture ≈ biggest piece + next biggest piece + … (k pieces)

WITH NUMBERS

k = 20: 28% of the numbers, only 4.9% off

Keeping only a few pieces is called a **low-rank approximation**.

LoRA: change a huge table with two thin strips



Next week you fine-tune a big model by learning only a small, low-rank change.

LoRA, as a formula

THE FORMULA

$$W' = W + BA$$

IN WORDS

new weights = old weights (kept frozen) + thin strip × thin strip

WITH NUMBERS

12,288 numbers to learn instead of 589,824: about 2%

PART 2

Surprise

How surprised is the model?

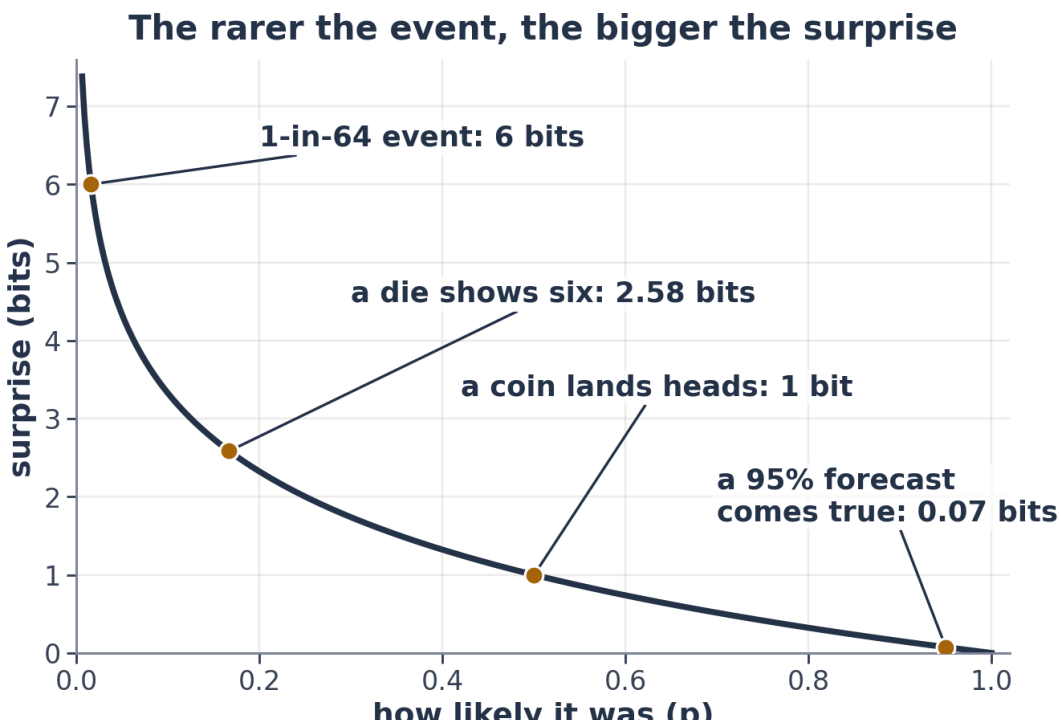
YOU ARE HERE

SHAPE
which way does the data stretch?

SURPRISE
how surprised is the model?

SPACE
where does real data live?

Rare things are surprising



A coin landing heads: a little surprise. A 1-in-64 event: a big one.

Surprise, as a formula

THE FORMULA

$$\text{surprise} = -\log_2 p$$

IN WORDS

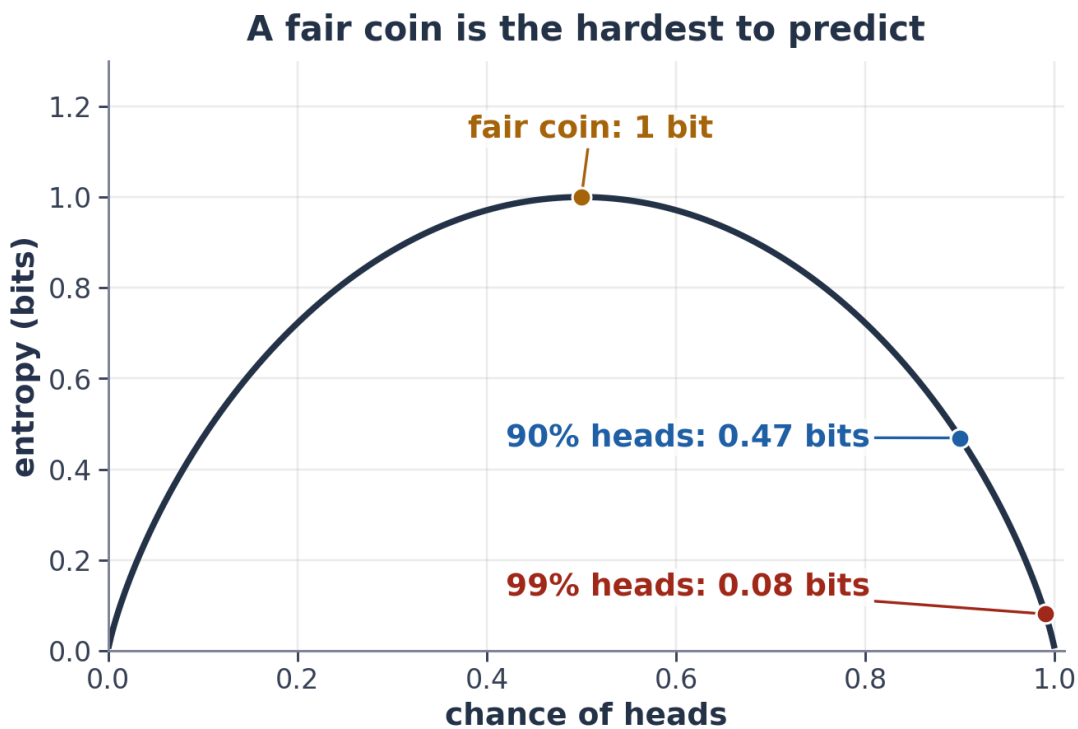
surprise = minus the log of how likely it was

WITH NUMBERS

a coin: $-\log_2(1/2) = 1$ bit · a 1-in-64 event: 6 bits

Measured in **bits**. PyTorch uses the natural log, giving **nats**: 1 nat ≈ 1.44 bits.

How unpredictable is a coin?



Entropy is the average surprise. A fair coin is the hardest to predict.

Entropy, as a formula

THE FORMULA

$$H = -\sum p \log_2 p$$

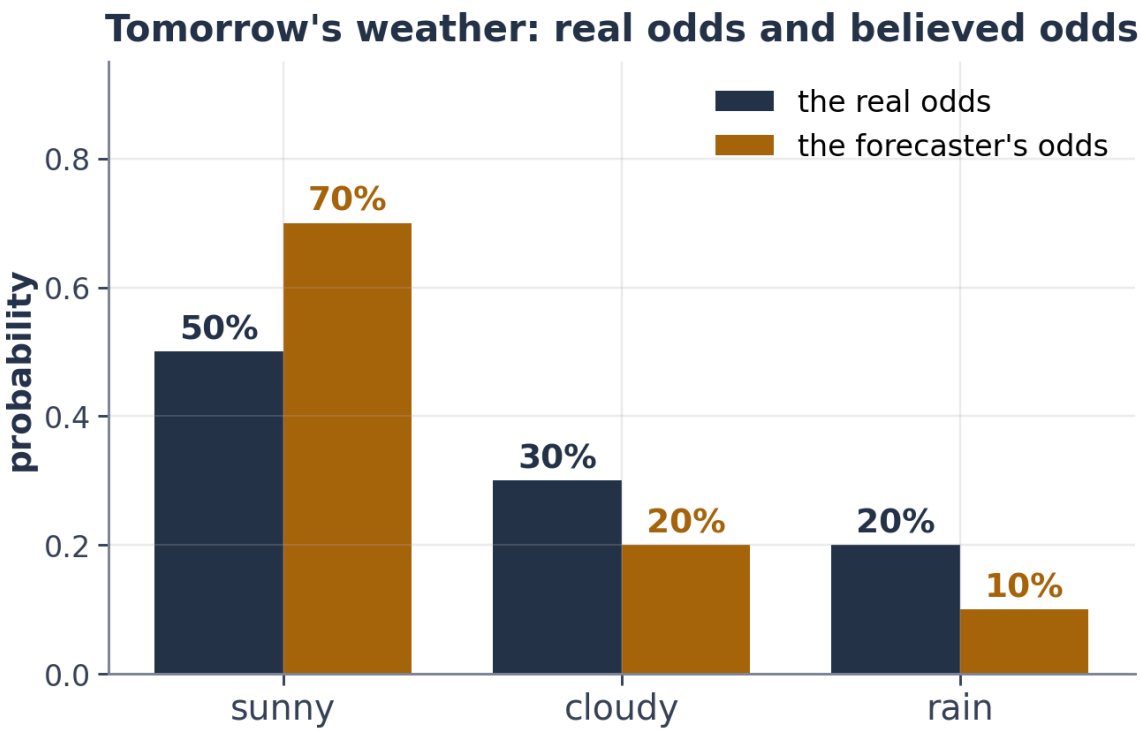
IN WORDS

entropy = the surprise of each outcome, averaged by how often it happens

WITH NUMBERS

fair coin: 1 bit · a 90/10 coin: 0.47 bits

A forecaster with the wrong odds



If you believe the wrong odds, real weather surprises you more than it needs to.

Cross-entropy, as a formula

THE FORMULA

$$H(p, q) = -\sum p \log_2 q$$

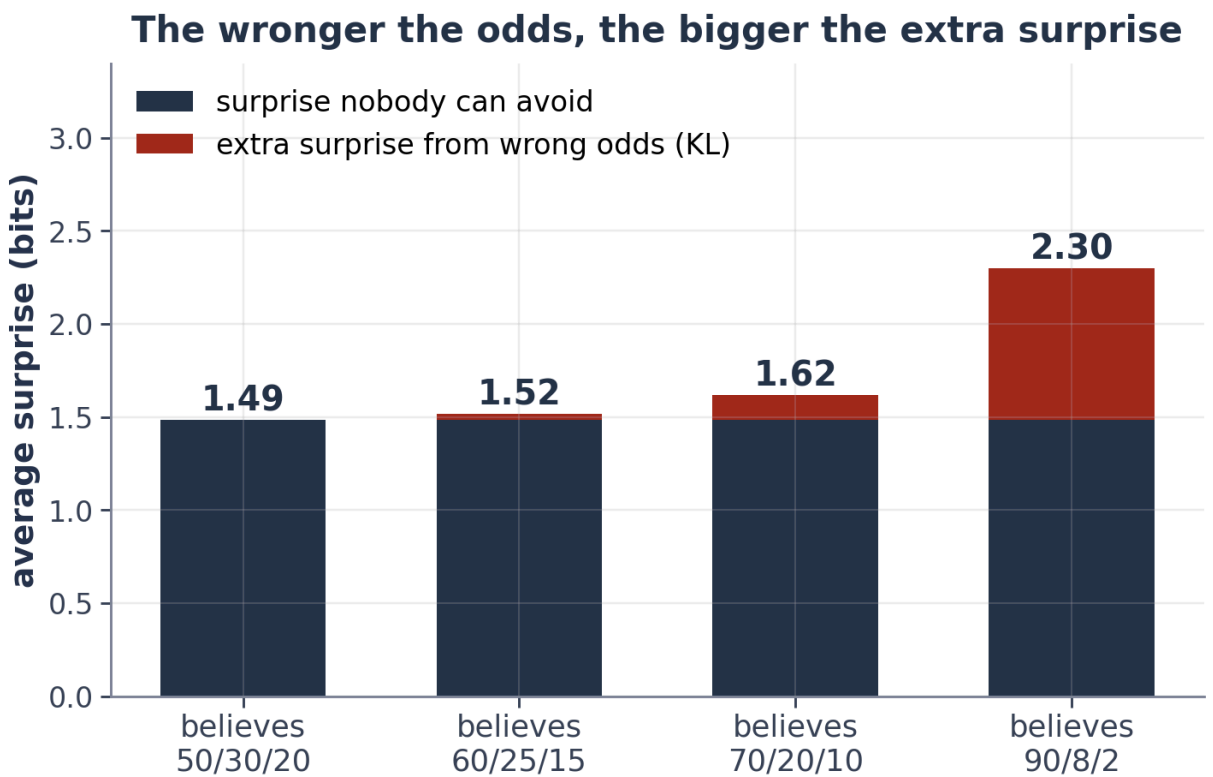
IN WORDS

cross-entropy = average surprise when the real odds are p but you believe q

WITH NUMBERS

believing 70/20/10 when the truth is 50/30/20: 1.62 bits instead of 1.49

Wrong odds cost extra surprise



KL divergence, as a formula

THE FORMULA

$$D_{\mathrm{KL}} = H(p, q) - H(p)$$

IN WORDS

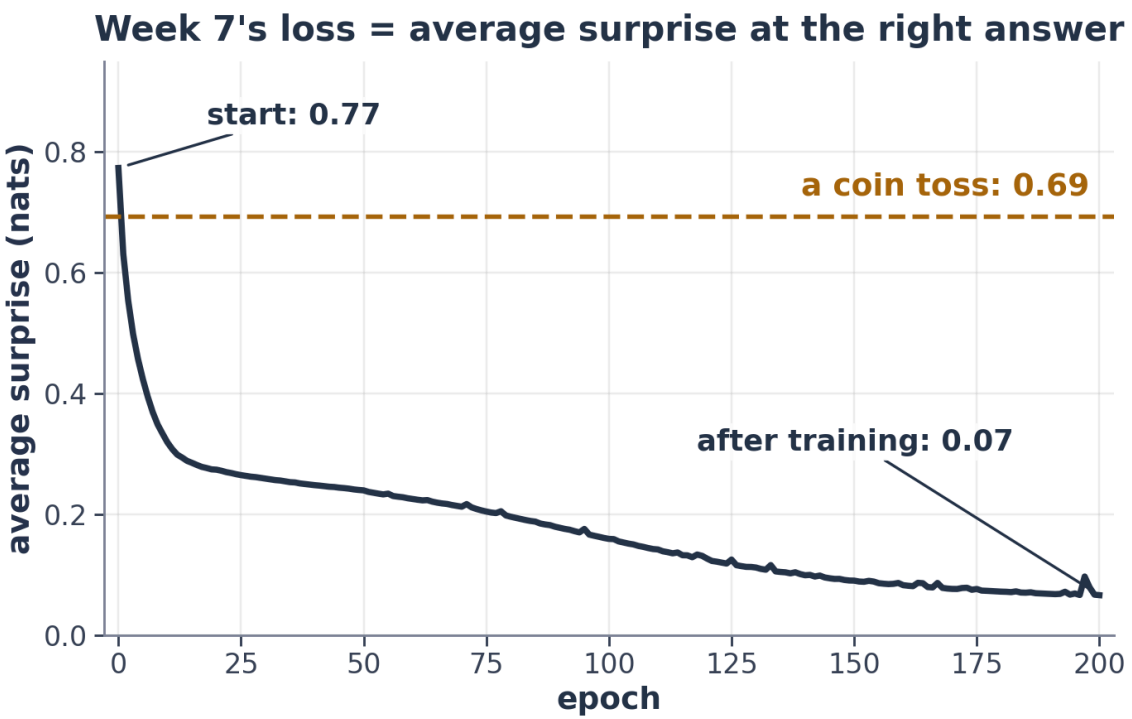
KL = the extra surprise you pay for believing the wrong odds

WITH NUMBERS

1.62 − 1.49 = 0.13 bits

KL is never below zero, and is zero only when you believe the real odds.

Your Week 7 loss was the average surprise



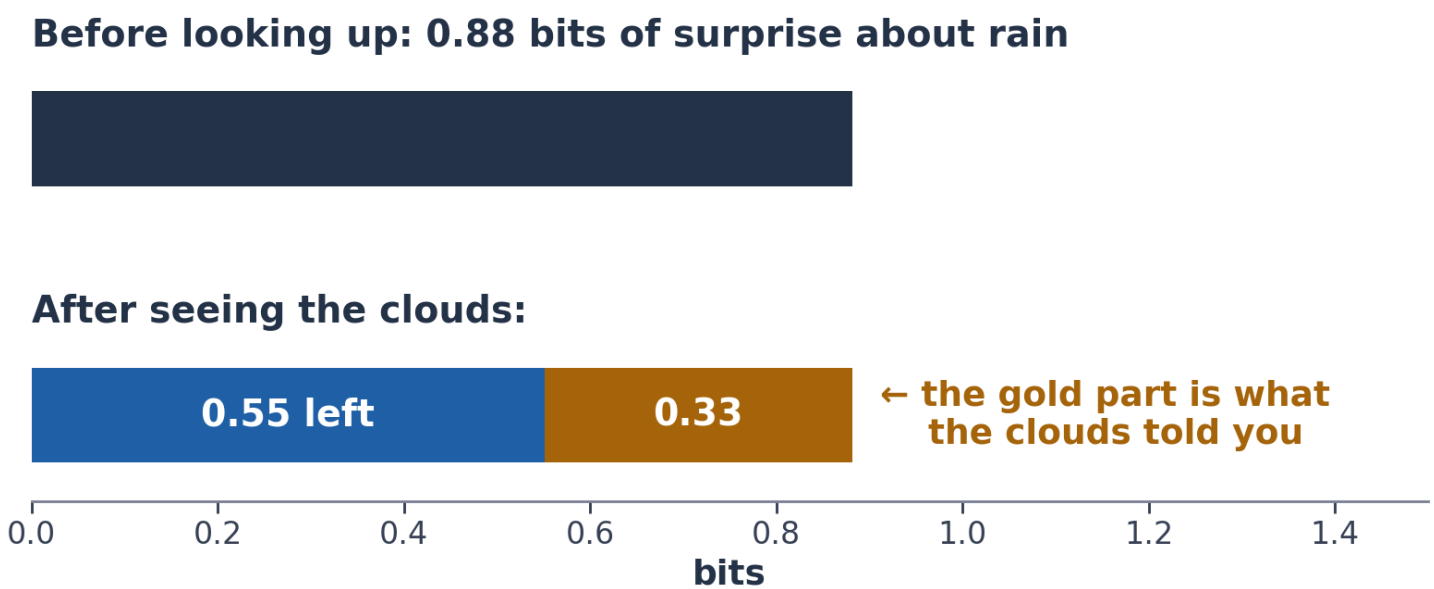
Training brought the surprise at the right answer down from 0.773 to 0.066.

Why cross-entropy is the right loss



Training can only shrink the avoidable part, so it pulls the model’s odds towards the real ones.

What do the clouds tell you about rain?



Rain on 30% of days; dark clouds on 90% of rainy days and 20% of dry ones.

Mutual information, as a formula

THE FORMULA

$$I = H(\text{before}) - H(\text{after})$$

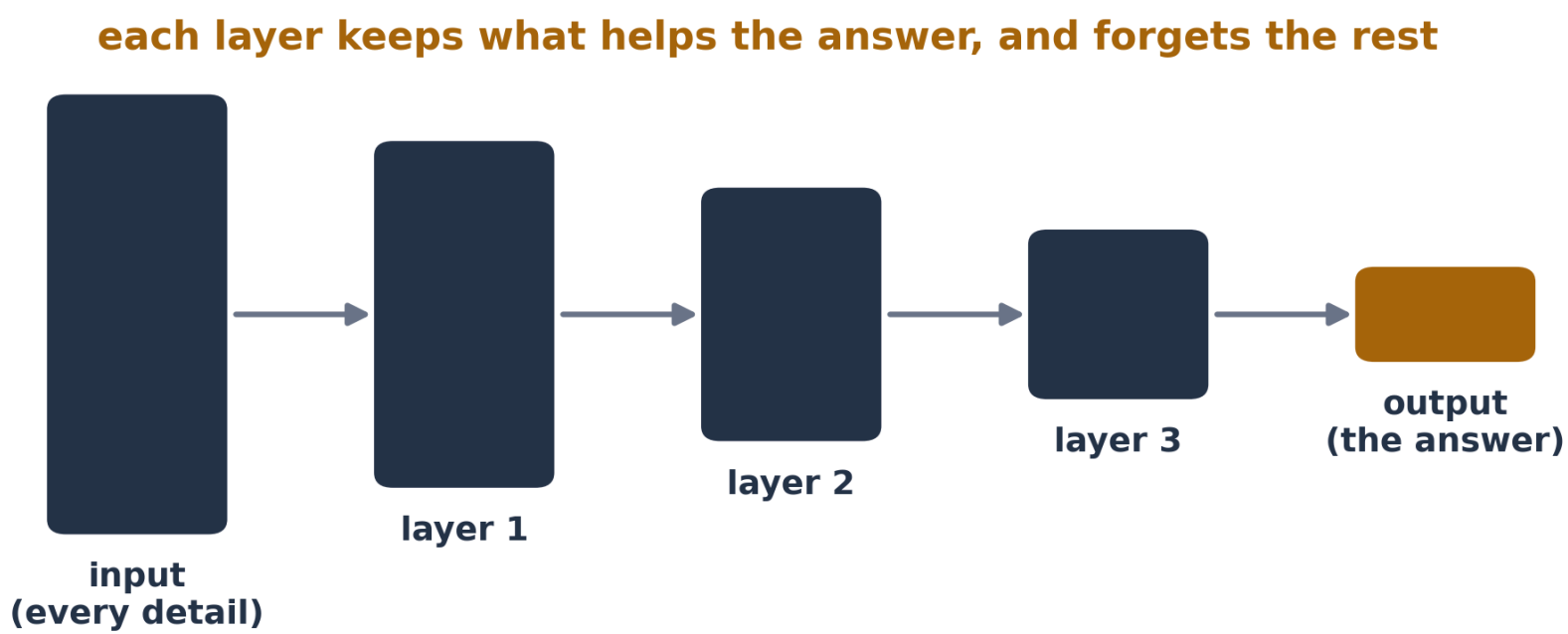
IN WORDS

what the clouds tell you = surprise about rain before looking − surprise after looking

WITH NUMBERS

0.881 − 0.551 = 0.33 bits

Each layer keeps what helps



A useful way to picture hidden layers: keep what helps the answer, forget the rest.

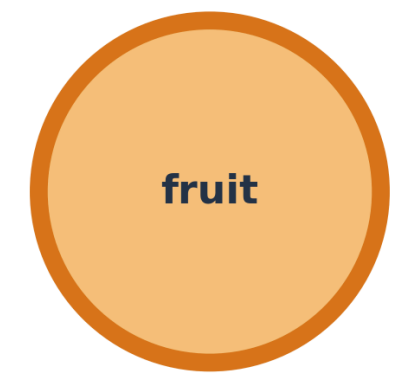
PART 3

Space

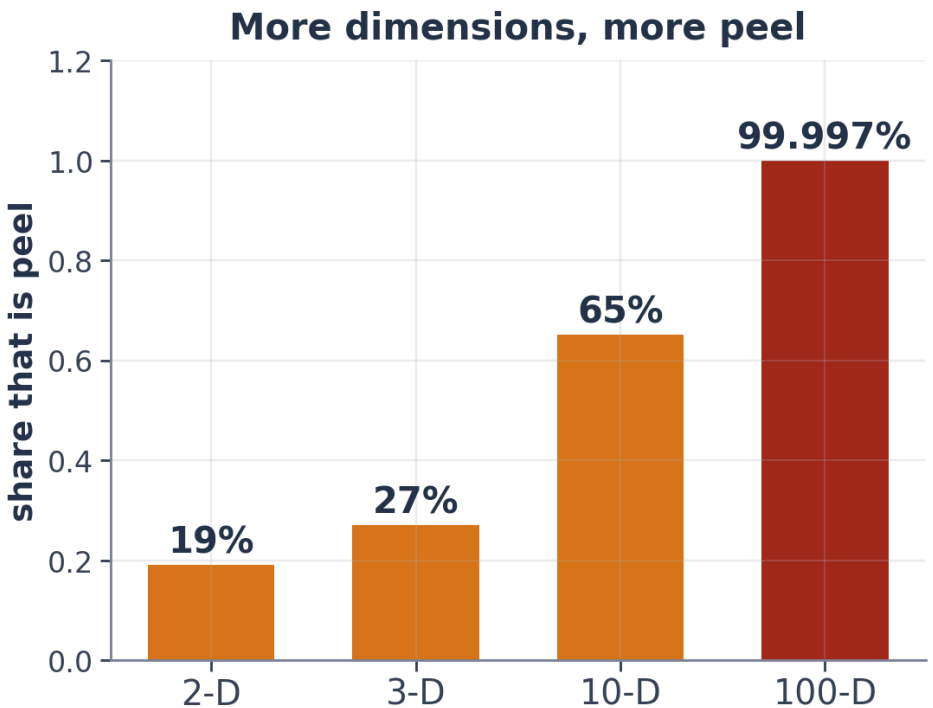
Where does real data live?

SHAPE	SURPRISE	YOU ARE HERE SPACE
which way does the data stretch?	how surprised is the model?	where does real data live?

In high dimensions, an orange is almost all peel



peel = the outer 10%



In many dimensions, almost every point is near the edge.

Volume near the edge, as a formula

THE FORMULA

$$\text{peel} = 1 - 0.9^{d}$$

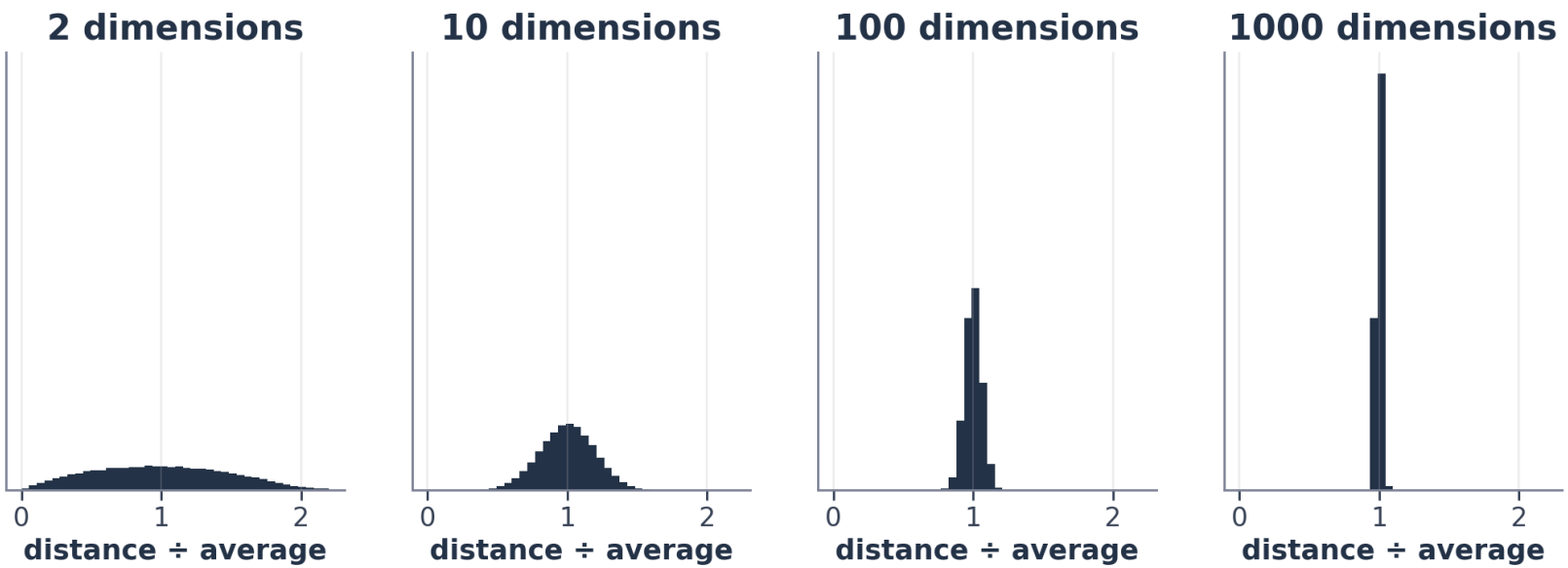
IN WORDS

peel = 1 – (0.9 × 0.9 × …, once for every dimension)

WITH NUMBERS

2-D: 19% · 10-D: 65% · 100-D: 99.997%

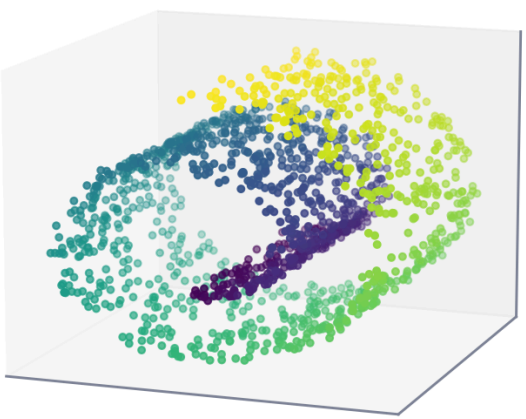
Everyone is about equally far away



In 1,000 dimensions the farthest pair is only 1.19× the nearest, so “nearest neighbour” stops meaning much.

Real data is a rolled-up sheet

3 numbers per point …

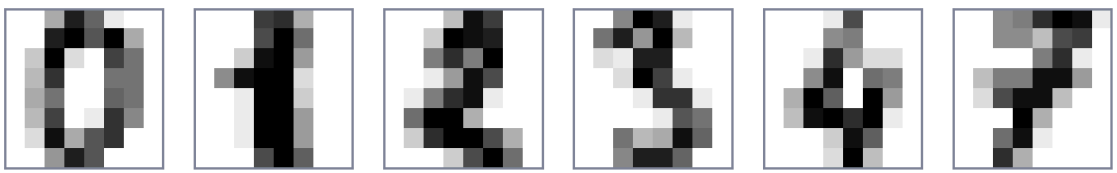


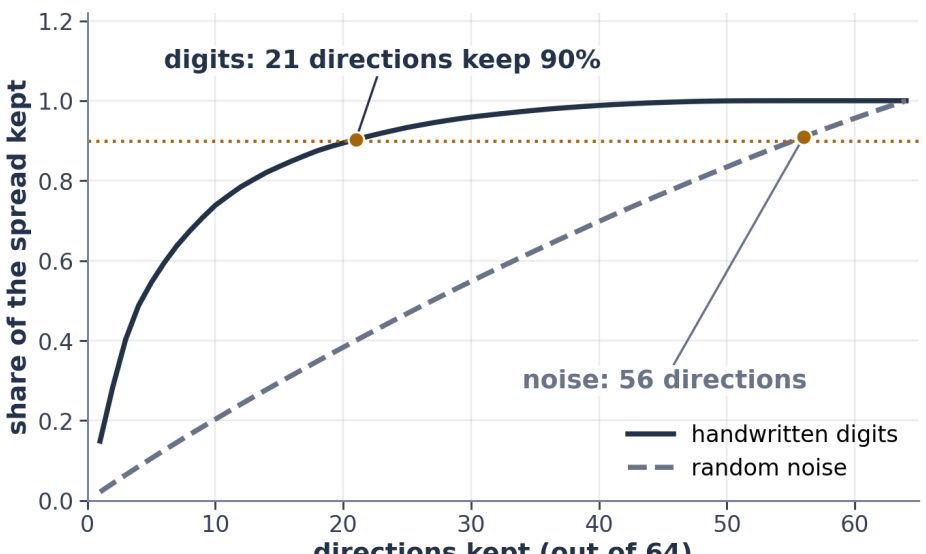
… but unrolled, a flat 2-D sheet



Like a sheet of paper rolled up in a room: 3 numbers per point, but really only 2.

Digits: 64 numbers, about 20 that matter





Counting the directions that matter

THE FORMULA

$$\text{smallest } k \text{ with } \frac{\sigma_1^2 + \ldots + \sigma_k^2}{\sigma_1^2 + \ldots + \sigma_n^2} \geq 0.9$$

IN WORDS

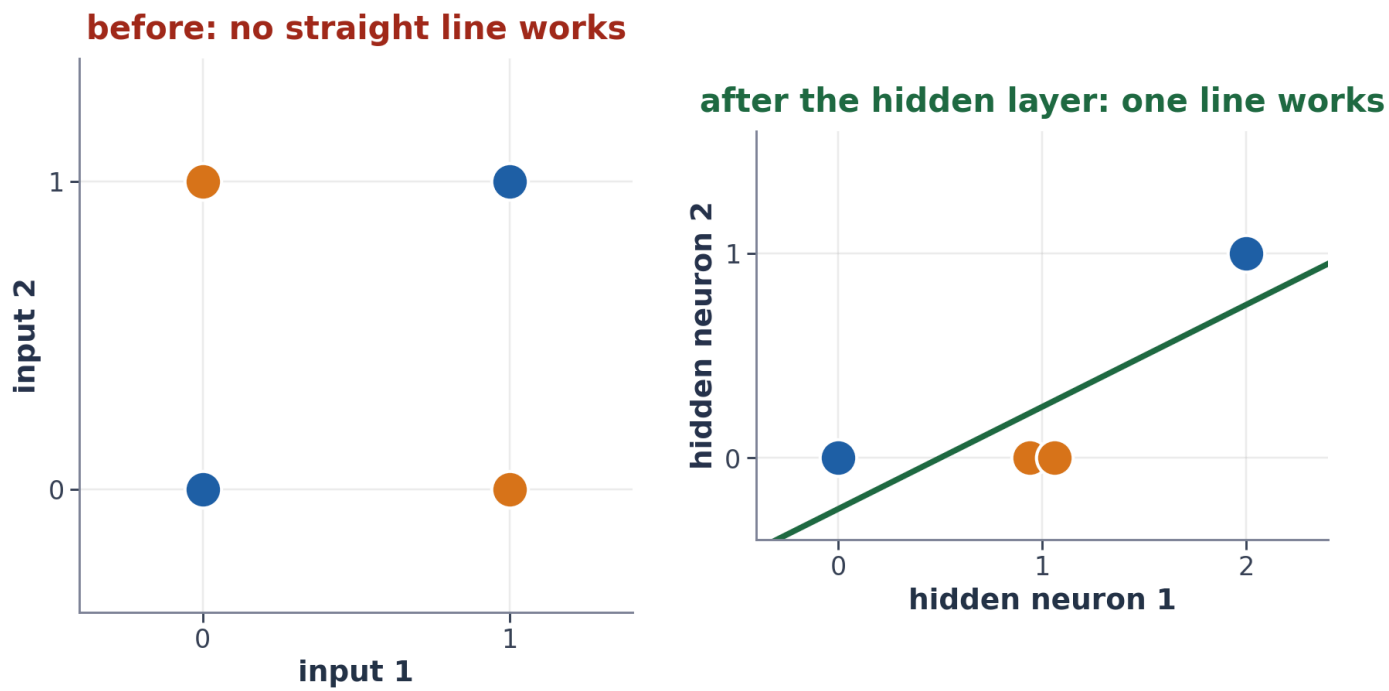
how many directions you need to keep 90% of the spread

WITH NUMBERS

digits: 21 of 64 · random noise: 56 of 64

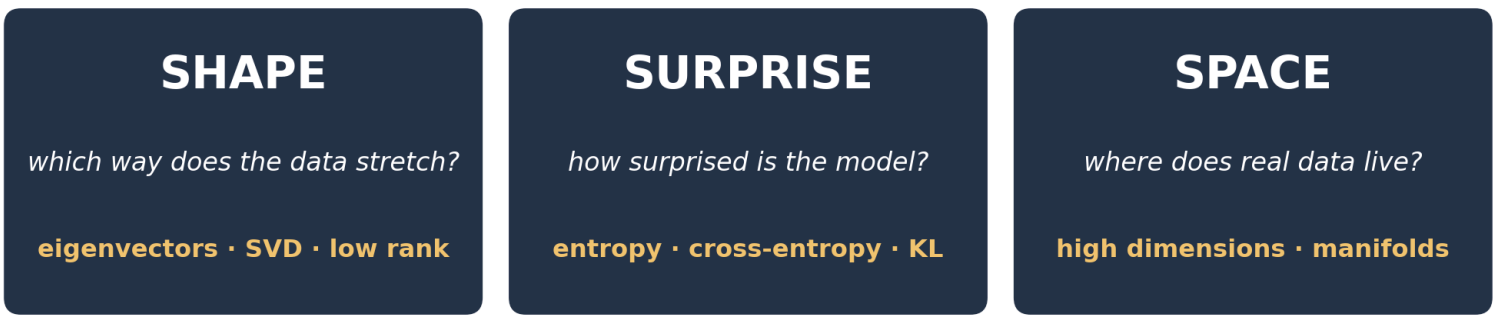
This rough count is the data's **intrinsic dimension**. You measure it for your own project data in the lab.

What a hidden layer does



It flattens the curved sheet the data lives on, until a straight line can cut it.

Shape, surprise, space



- **Shape:** the SVD finds the directions that matter; keeping a few is a low-rank approximation.
- **Surprise:** your loss is unavoidable surprise plus avoidable surprise (KL), and training shrinks the avoidable part.
- **Space:** real data fills only a thin, low-dimensional part of the space it is written in.

Where these ideas come back

Week 9 **LoRA**: change a big model with low-rank strips.

Week 10 Transformers: their outputs are odds, scored with **cross-entropy**.

Week 11 Diffusion: walk noisy points back onto the **rolled-up sheet** of real data.

Your capstone project starts now

1. Read the brief and the six elective one-pagers on Tutors.
2. Create your private repository from the Project Starter.
3. Submit the **Capstone Proposal Form** by **23:59 on Friday 6 November**.

Your data, or your personal variant, arrives on Monday 9 November with your reviewed proposal.

The Math Behind the Models

1. **PCA via SVD**, written by you and checked against scikit-learn; then shrink a photo with it. It becomes `src/scratch/pca_svd.py` in your capstone.
2. **Entropy, cross-entropy and KL** on a real model's outputs, and how they match its loss.
3. **How many directions matter** in your own project data.

Not graded: submit your SUBMISSION.md, then compare with the reference answers.

SHAPE
which way does the data stretch?

SURPRISE
how surprised is the model?

SPACE
where does real data live?

Shape. Surprise. Space.

Questions?